

Chapter 39

Plant Responses to Internal and External Signals

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

Neil Campbell and Jane Reece

Lectures by Chris Romero, updated by Erin Barley with contributions from Joan Sharp

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Overview: Stimuli and a Stationary Life

- Linnaeus noted that flowers of different species opened at different times of day and could be used as a *horologium florum*, or floral clock.
- Plants, being rooted to the ground, must respond to environmental changes that come their way.
- For example, the bending of a seedling toward light begins with sensing the direction, quantity, and color of the light.

Signal transduction pathways link signal reception to response

- Plants have cellular receptors that detect changes in their environment.
- For a stimulus to elicit a response, certain cells must have an appropriate receptor.
- *Stimulation of the receptor initiates a specific signal transduction pathway.*

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- A potato left growing in darkness produces shoots that look unhealthy and lacks elongated roots. These are morphological adaptations for growing in darkness, collectively called **etiolation**.
 - After exposure to light, a potato undergoes changes called **de-etiolation**, in which shoots and roots grow normally.
 - A potato's response to light is *cell-signal processing*: The stages are *reception, transduction, and response*.

Light-induced de-etiolation (greening) of dark-grown potatoes

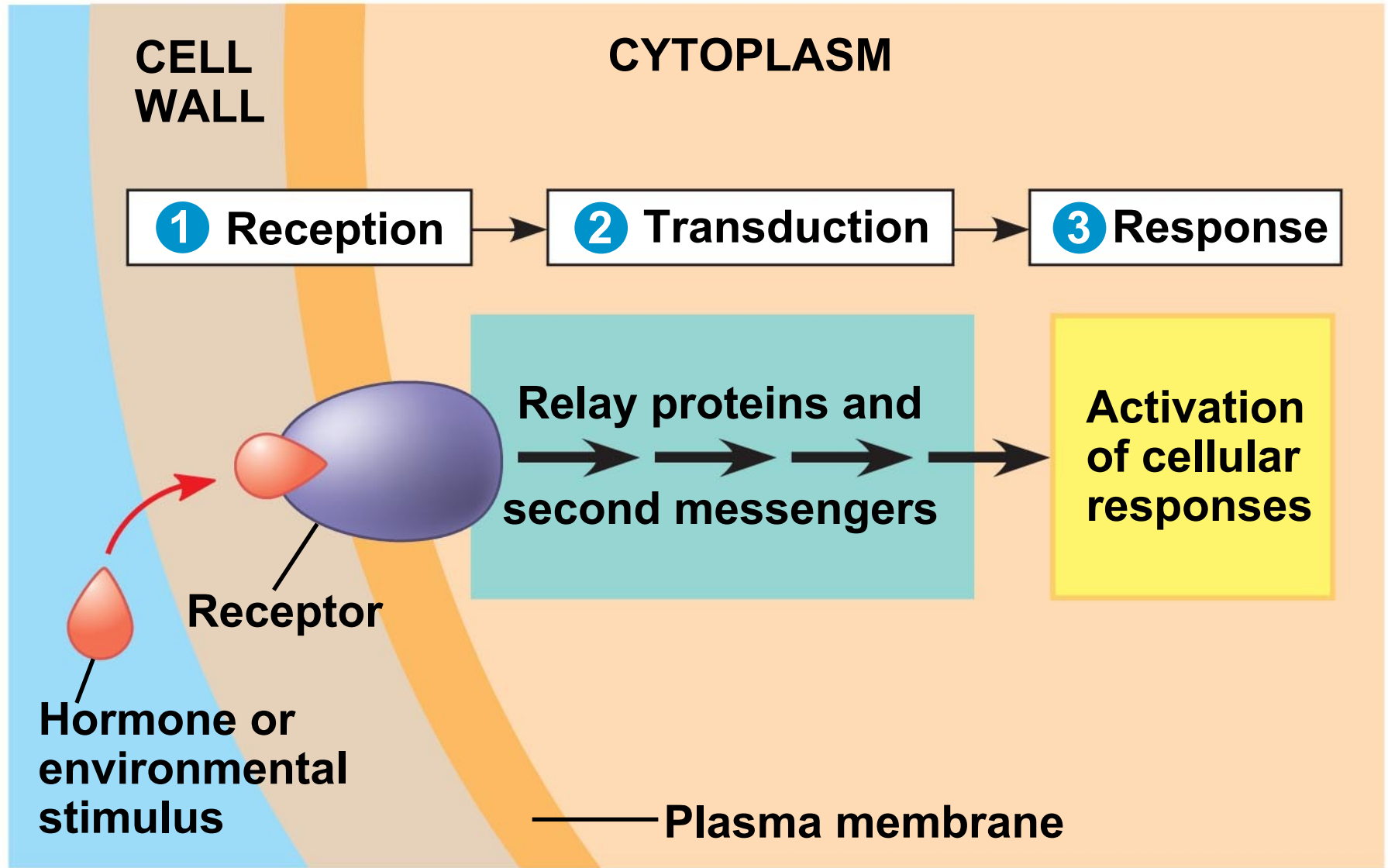


(a) Before exposure to light



(b) After a week's exposure to natural daylight

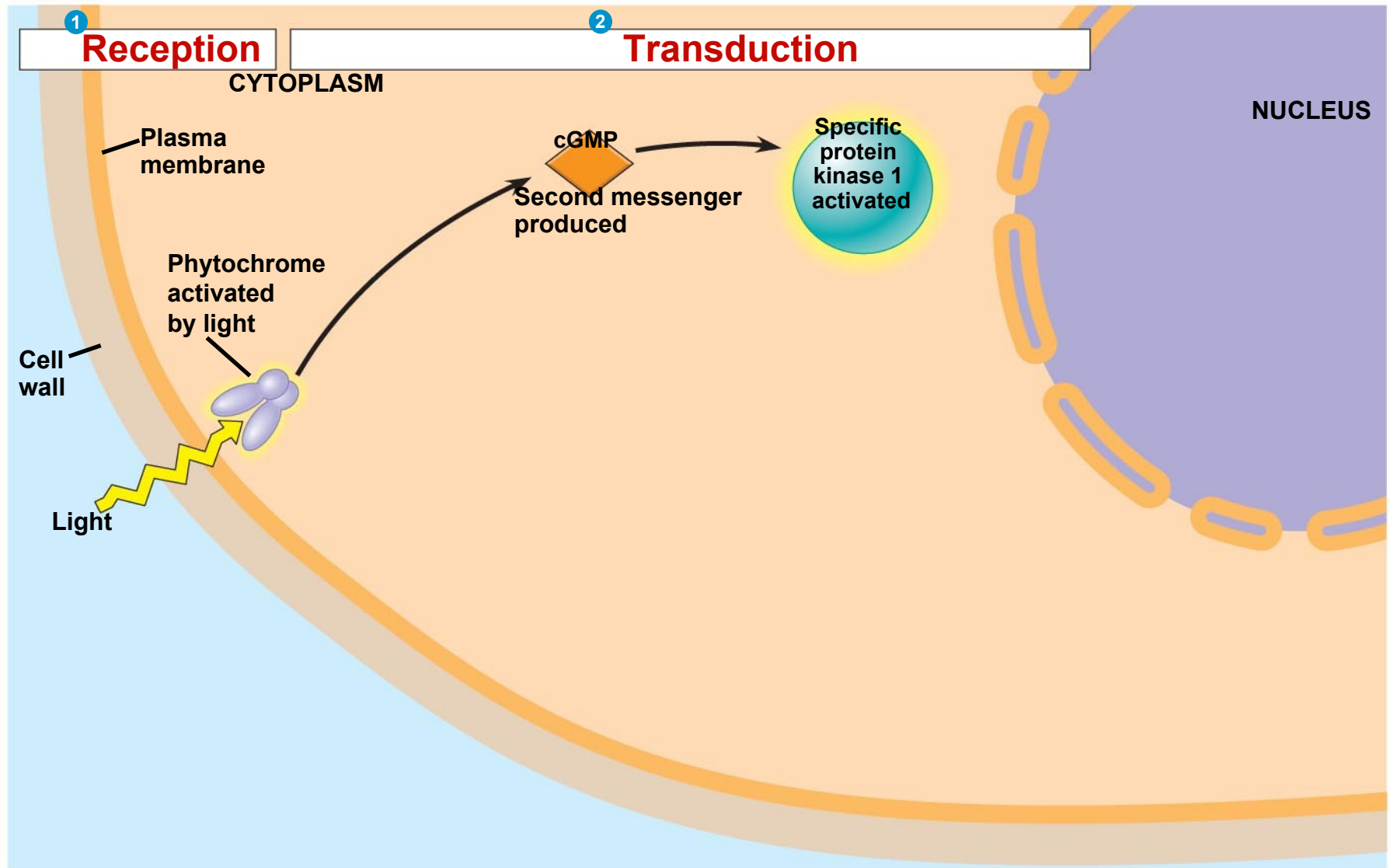
Signal Transduction Pathways



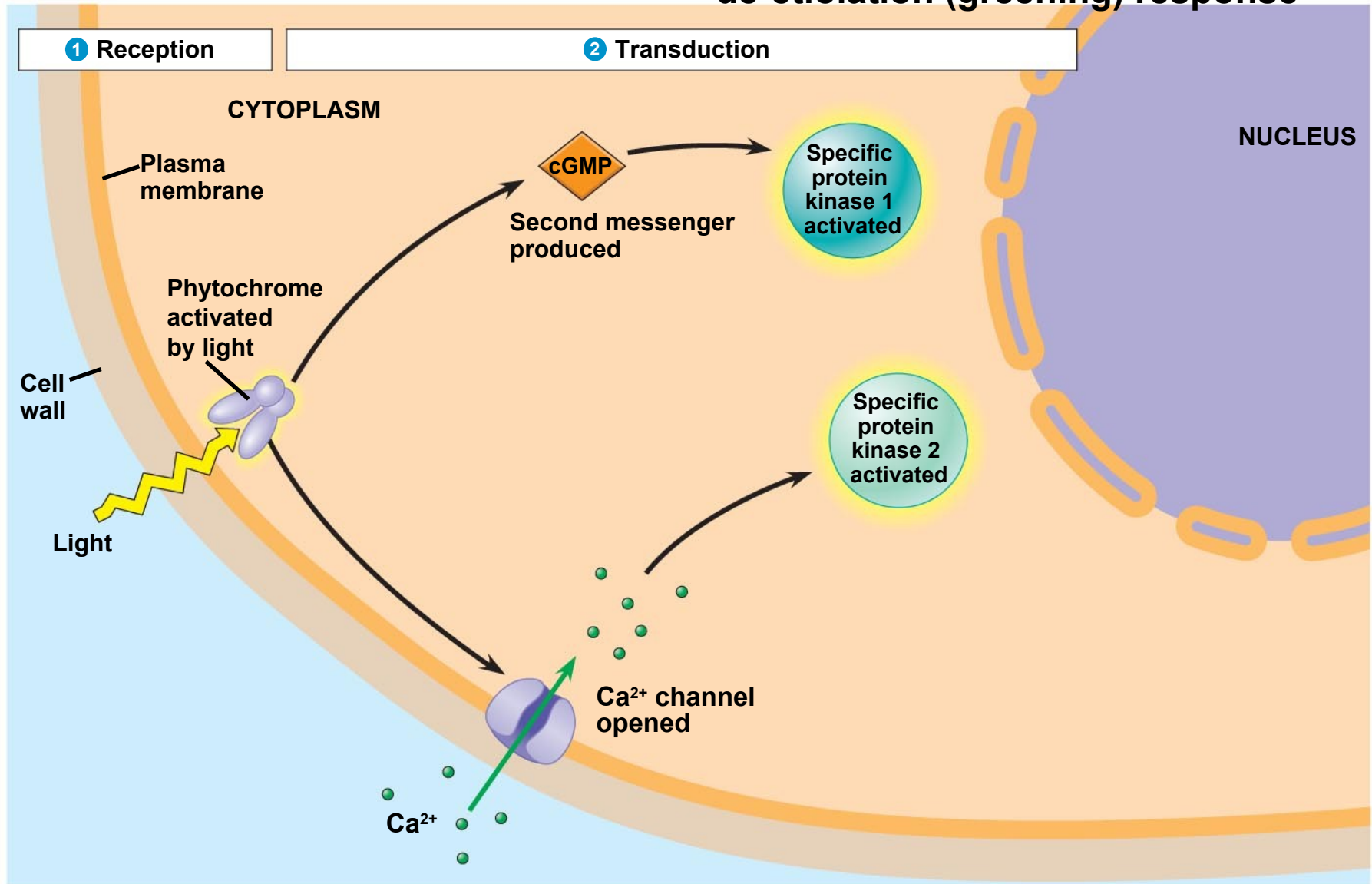
Reception and Transduction

- **Receptors** - Internal and external signals are detected by *shape match* receptors, *proteins that change in response to specific stimuli*.
- **Second messengers** transfer and amplify signals from receptors to proteins that cause responses.

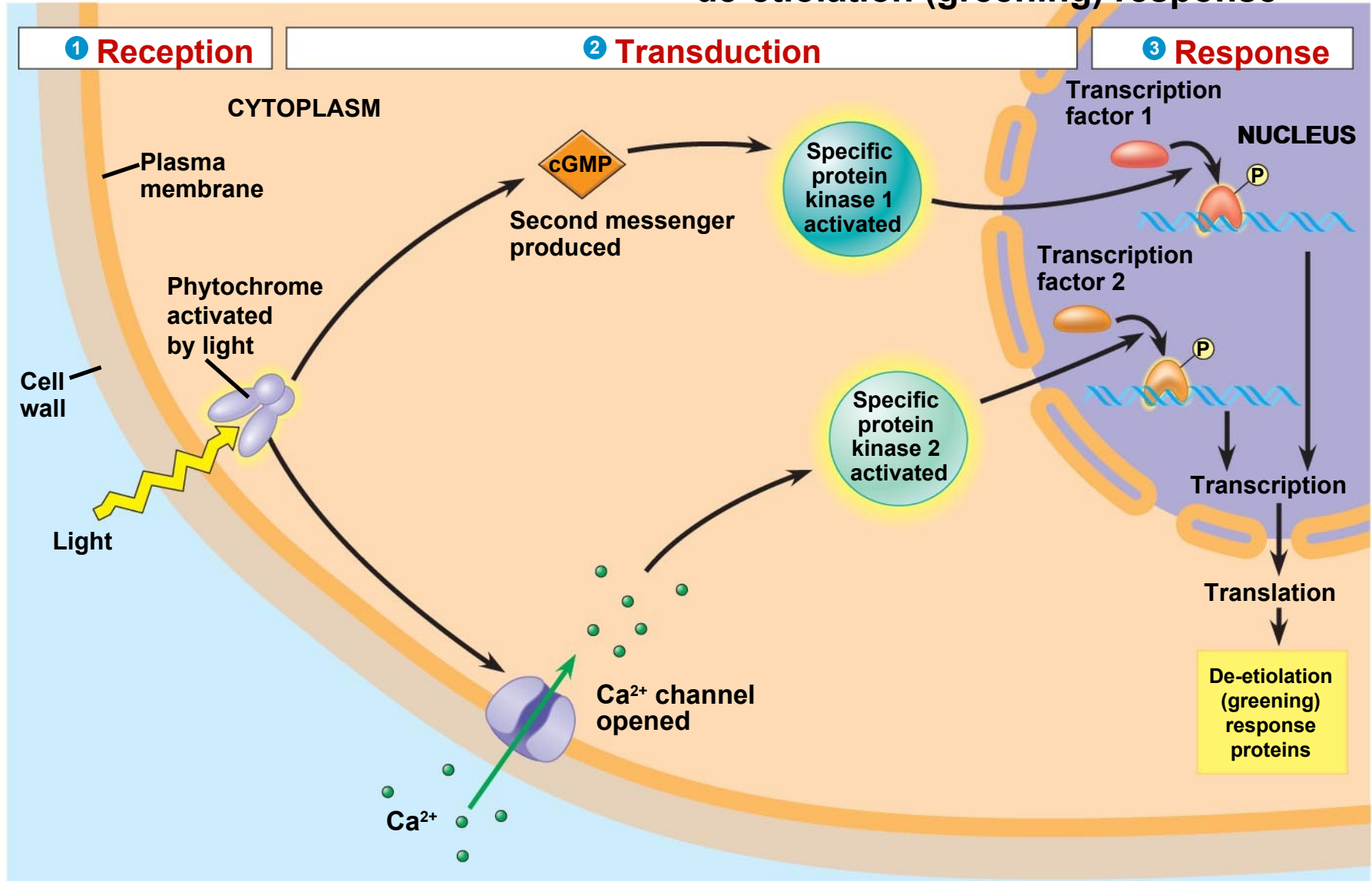
Signal Transduction in plants: the role of phytochrome in the de-etiolation (greening) response



Signal Transduction in plants: the role of phytochrome in the de-etiolation (greening) response



Signal Transduction in plants: the role of phytochrome in the de-etiolation (greening) response



Response

Response - A signal transduction pathway leads to **regulation of one or more cellular activities**.

- In most cases, these responses to stimulation involve *increased activity of enzymes*.
- This can occur by *transcriptional regulation or post-translational modification*.

Transcriptional Regulation

- *Specific transcription factors* bind directly to specific regions of DNA and *control transcription of genes*.
- Positive transcription factors are proteins that *increase* the transcription of specific genes, while negative transcription factors are proteins that *decrease* the transcription of specific genes.

Post-Translational Modification of Proteins

- *Post-translational modification* involves modification of existing proteins in the signal response.
- Modification *often involves the phosphorylation of specific amino acids.*

De-Etiolation (“Greening”) Proteins

- Many enzymes that function in certain signal responses are directly involved in photosynthesis.
- Other enzymes are involved in supplying chemical precursors for chlorophyll production.

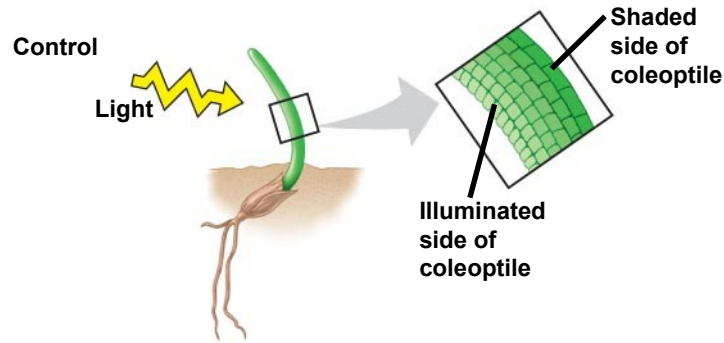
Plant hormones help coordinate growth, development, and responses to stimuli

- **Hormones** are chemical signals that coordinate different parts of an organism.
- Any response resulting in curvature of organs *toward or away from a stimulus* is called a ***tropism = a growth response***.
- Tropisms are often caused by hormones.

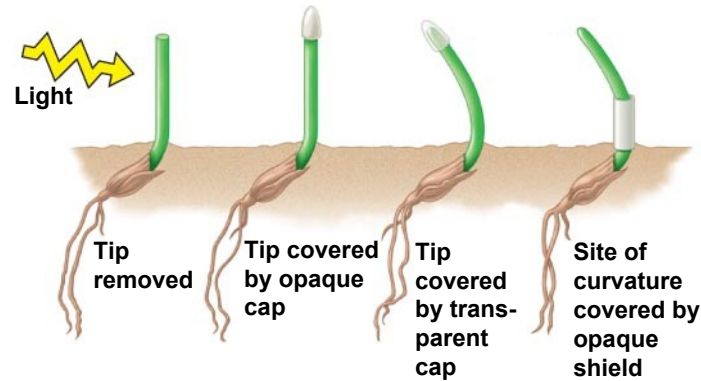
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- In the late 1800s, Charles Darwin and his son Francis conducted experiments on **phototropism**, a plant's response to light.
 - They observed that a grass seedling could bend toward light only if the tip of the coleoptile was present.
 - They postulated that a signal was transmitted from the tip to the elongating region.

Signaling and Phototropism

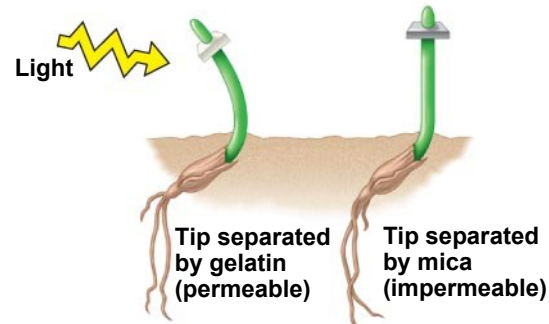
RESULTS



Phototropic response only when tip is illuminated



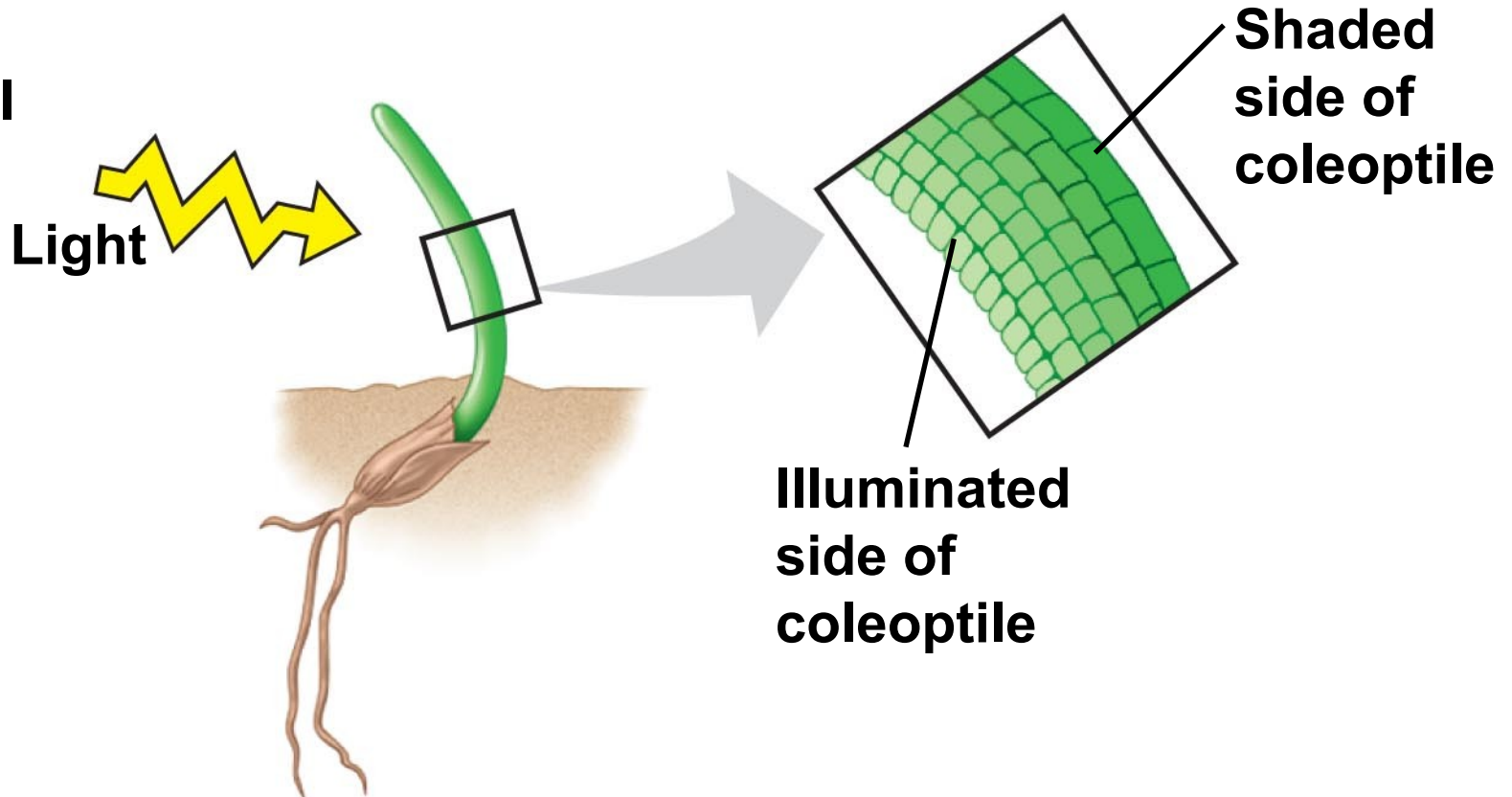
Phototropic response when tip separated by permeable barrier, but not with impermeable barrier



What part of a grass coleoptile senses light, and how is the signal transmitted?

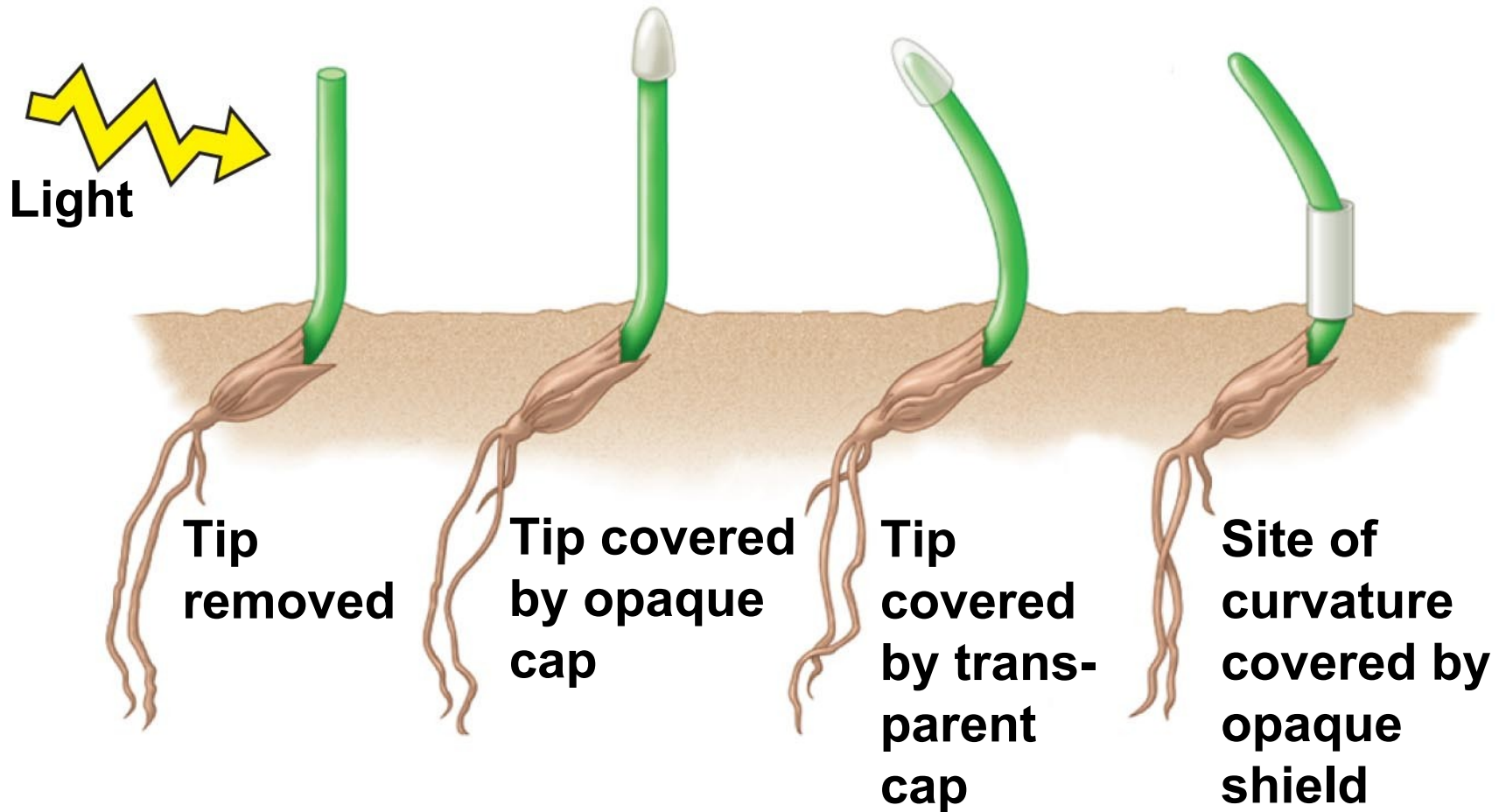
RESULTS

Control



RESULTS

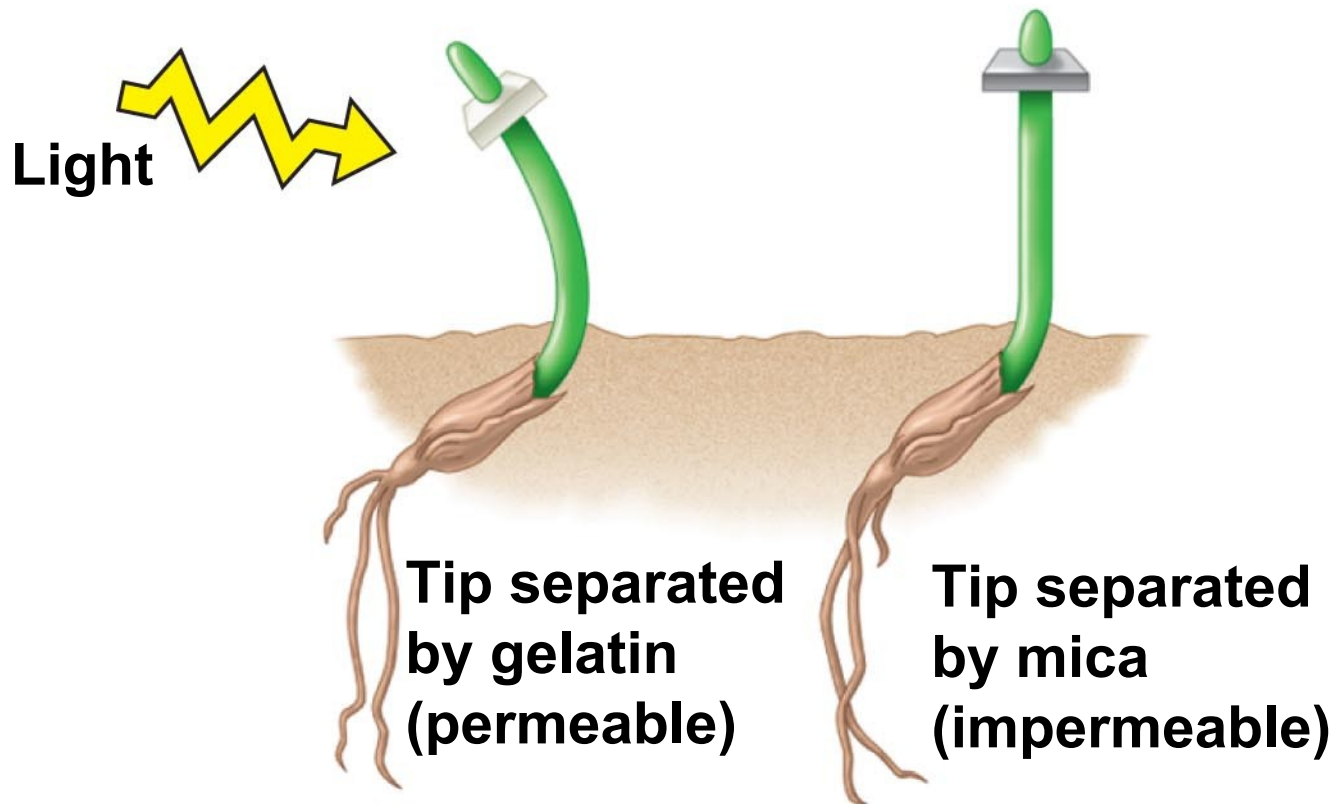
Phototropic response only when tip is illuminated



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- In 1913, Peter Boysen-Jensen demonstrated that the signal was a mobile chemical substance.
 - In 1926, Frits Went extracted the chemical messenger for phototropism, auxin, by modifying earlier experiments.

RESULTS

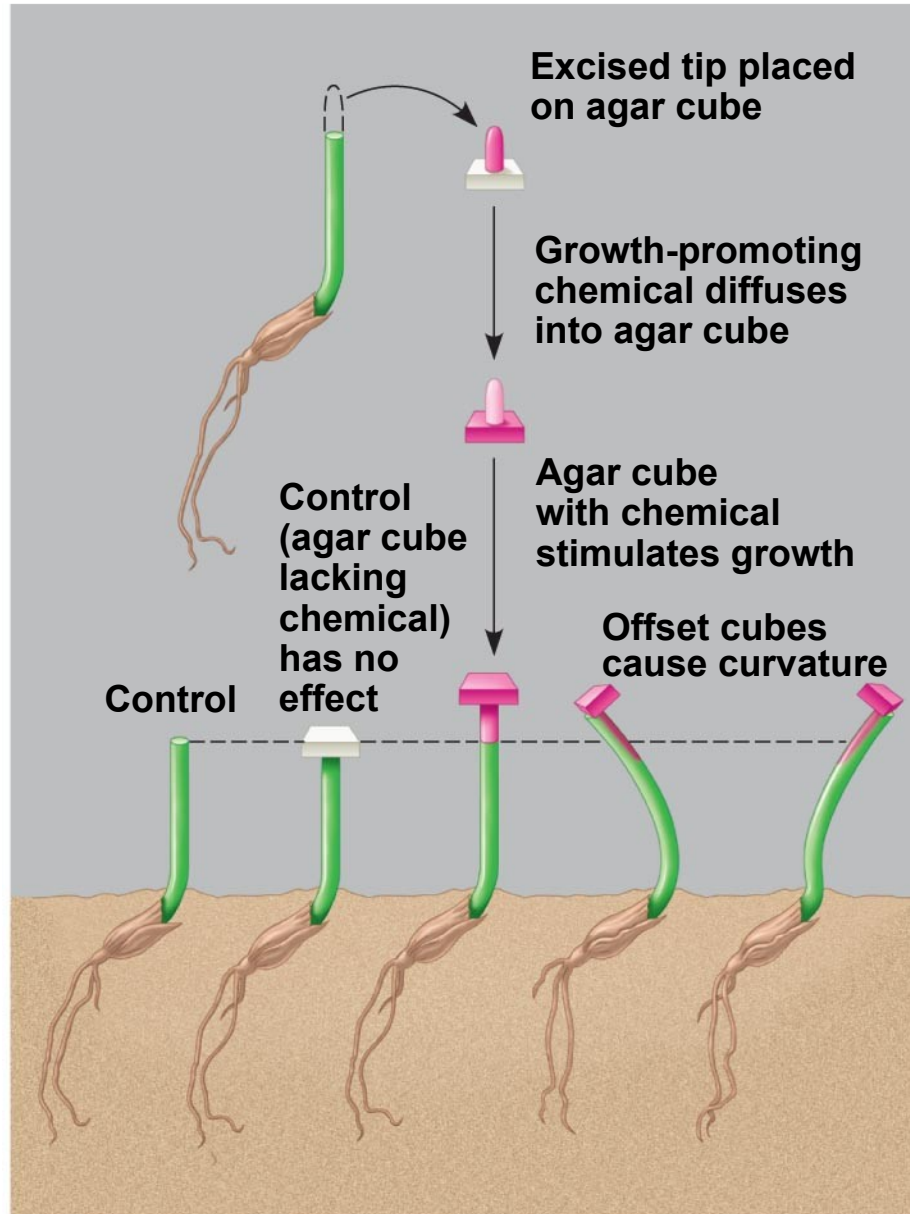
Boysen-Jensen: phototropic response when tip is separated by permeable barrier, but not with impermeable barrier



Question:

Does asymmetric distribution of a growth-promoting chemical cause a coleoptile to grow toward the light?

RESULTS



A Survey of Plant Hormones

- In general, ***hormones control plant growth and development*** by affecting the ***division, elongation, and differentiation of cells***.
- Plant hormones are produced in very low concentration, but a minute amount can greatly affect growth and development of a plant organ.

Table 39.1 Overview of Plant Hormones

Hormone	Where Produced or Found in Plant	Major Functions
Auxin (IAA)	Shoot apical meristems and young leaves are the primary sites of auxin synthesis. Root apical meristems also produce auxin, although the root depends on the shoot for much of its auxin. Developing seeds and fruits contain high levels of auxin, but it is unclear whether it is newly synthesized or transported from maternal tissues.	Stimulates stem elongation (low concentration only); promotes the formation of lateral and adventitious roots; regulates development of fruit; enhances apical dominance; functions in phototropism and gravitropism; promotes vascular differentiation; retards leaf abscission.
Cytokinins	These are synthesized primarily in roots and transported to other organs, although there are many minor sites of production as well.	Regulate cell division in shoots and roots; modify apical dominance and promote lateral bud growth; promote movement of nutrients into sink tissues; stimulate seed germination; delay leaf senescence.
Gibberellins	Meristems of apical buds and roots, young leaves, and developing seeds are the primary sites of production.	Stimulate stem elongation, pollen development, pollen tube growth, fruit growth, and seed development and germination; regulate sex determination and the transition from juvenile to adult phases.
Brassinosteroids	These compounds are present in all plant tissues, although different intermediates predominate in different organs. Internally produced brassinosteroids act near the site of synthesis.	Promote cell expansion and cell division in shoots; promote root growth at low concentrations; inhibit root growth at high concentrations; promote xylem differentiation and inhibit phloem differentiation; promote seed germination and pollen tube elongation.
Absciscic acid (ABA)	Almost all plant cells have the ability to synthesize absciscic acid, and its presence has been detected in every major organ and living tissue; may be transported in the phloem or xylem.	Inhibits growth; promotes stomatal closure during drought stress; promotes seed dormancy and inhibits early germination; promotes leaf senescence; promotes desiccation tolerance.
Ethylene	This gaseous hormone can be produced by almost all parts of the plant. It is produced in high concentrations during senescence, leaf abscission, and the ripening of some types of fruits. Synthesis is also stimulated by wounding and stress.	Promotes ripening of many types of fruit, leaf abscission, and the triple response in seedlings (inhibition of stem elongation, promotion of lateral expansion, and horizontal growth); enhances the rate of senescence; promotes root and root hair formation; promotes flowering in the pineapple family.

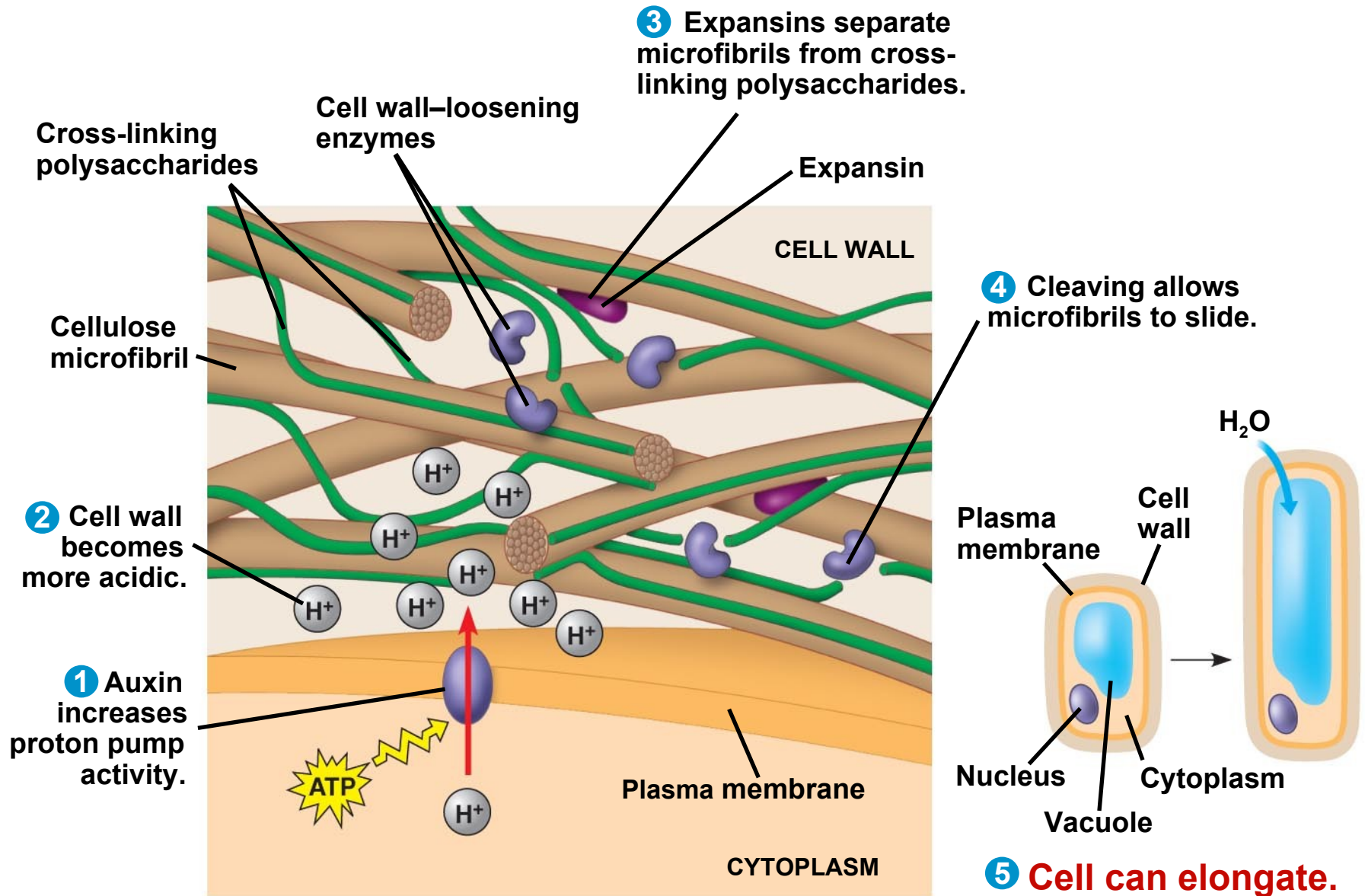
Auxin - Polar Movement of Auxins

- The term **auxin** refers to any chemical that promotes **elongation** of coleoptiles.
- **Indoleacetic acid** (IAA) is a common auxin in plants; in this lecture the term auxin refers specifically to **IAA**.
- Auxin **transporter proteins** move the hormone from the basal end of one cell into the apical end of the neighboring cell.

The Role of Auxin in **Cell Elongation**

- According to the *acid growth hypothesis*, auxin stimulates proton pumps in the plasma membrane.
- The *proton pumps* H^+ lower the pH in the cell wall, activating **expansins**, enzymes that *loosen the cell wall's fabric*.
- With the cellulose loosened, the cell can elongate.

Cell elongation in response to auxin: acid growth hypothesis



Auxins

Lateral and Adventitious Root Formation

- Auxin is involved in root formation and branching

Auxins as Herbicides

- An overdose of synthetic auxins can kill eudicots

Other Effects of Auxin

- Auxin affects secondary growth by inducing cell division in the vascular cambium and influencing differentiation of secondary xylem

Cytokinins

Cytokinins are so named because they stimulate cytokinesis (cell division).

Control of Cell Division and Differentiation

- Cytokinins are produced in actively growing tissues such as roots, embryos, and fruits.
- Cytokinins work together with auxin to control cell division and differentiation.

Cytokinins

Control of Apical Dominance

- Cytokinins, auxin, and other factors interact in the control of apical dominance, a terminal bud's ability to suppress development of axillary buds
- If the terminal bud is removed, plants become bushier.

Anti-Aging Effects

- Cytokinins retard the aging of some plant organs by inhibiting protein breakdown, stimulating RNA and protein synthesis, and mobilizing nutrients from surrounding tissues.

Gibberellins

Gibberellins have a variety of effects, such as stem elongation, fruit growth, and seed germination.

Stem Elongation

- Gibberellins stimulate growth of leaves and stems.
- In stems, they stimulate cell elongation and cell division.

Fruit Growth

- In many plants, both auxin and gibberellins must be present for fruit to set.
- Gibberellins are used in spraying of Thompson seedless grapes.

Germination

- After water is imbibed, release of gibberellins from the embryo signals seeds to germinate.

Effects of gibberellins on stem elongation and fruit growth



(a) Gibberellin-induced stem growth



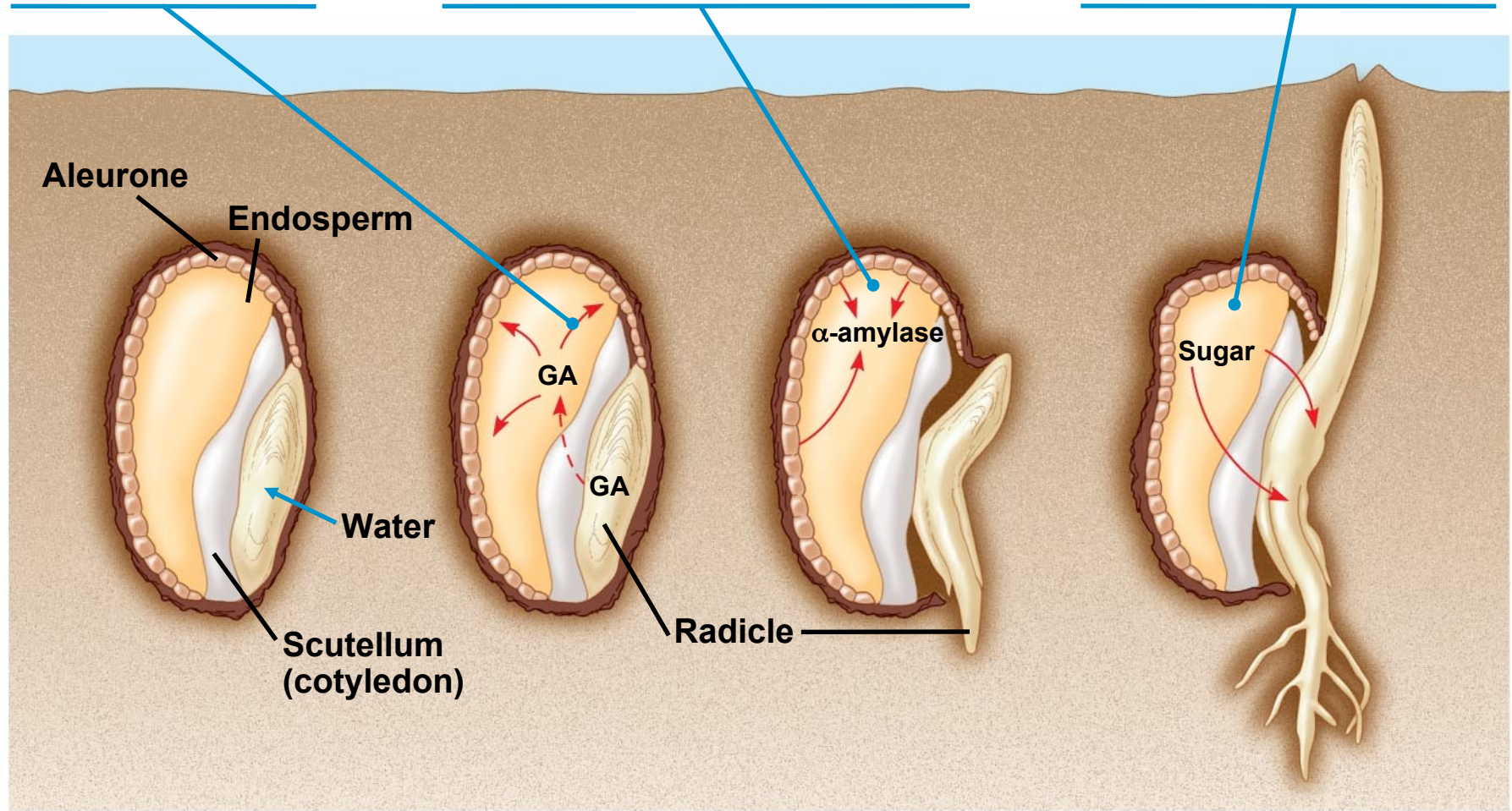
(b) Gibberellin-induced fruit growth

Mobilization of nutrients by gibberellins during the germination of seeds such as barley

1 Gibberellins (GA) send signal to aleurone.

2 Aleurone secretes α -amylase and other enzymes.

3 Sugars and other nutrients are consumed.



Brassinosteroids

- **Brassinosteroids** are chemically similar to the sex hormones of animals.
- They induce cell elongation and division in stem segments.

Abscisic Acid

- **Abscisic acid (ABA)** slows growth.
- Two of the many effects of ABA:
 - Seed dormancy
 - Drought tolerance.

Abscisic Acid = ABA

Seed Dormancy

- Seed dormancy ensures that the seed will germinate only in optimal conditions.
- In some seeds, dormancy is broken when ABA is removed by heavy rain, light, or prolonged cold.

Drought Tolerance

- ABA is the primary internal signal that enables plants to withstand drought.

Ethylene

- Plants produce **ethylene** in response to stresses such as drought, flooding, mechanical pressure, injury, and infection.
- The effects of ethylene include response to fruit ripening, mechanical stress, senescence, and leaf abscission.

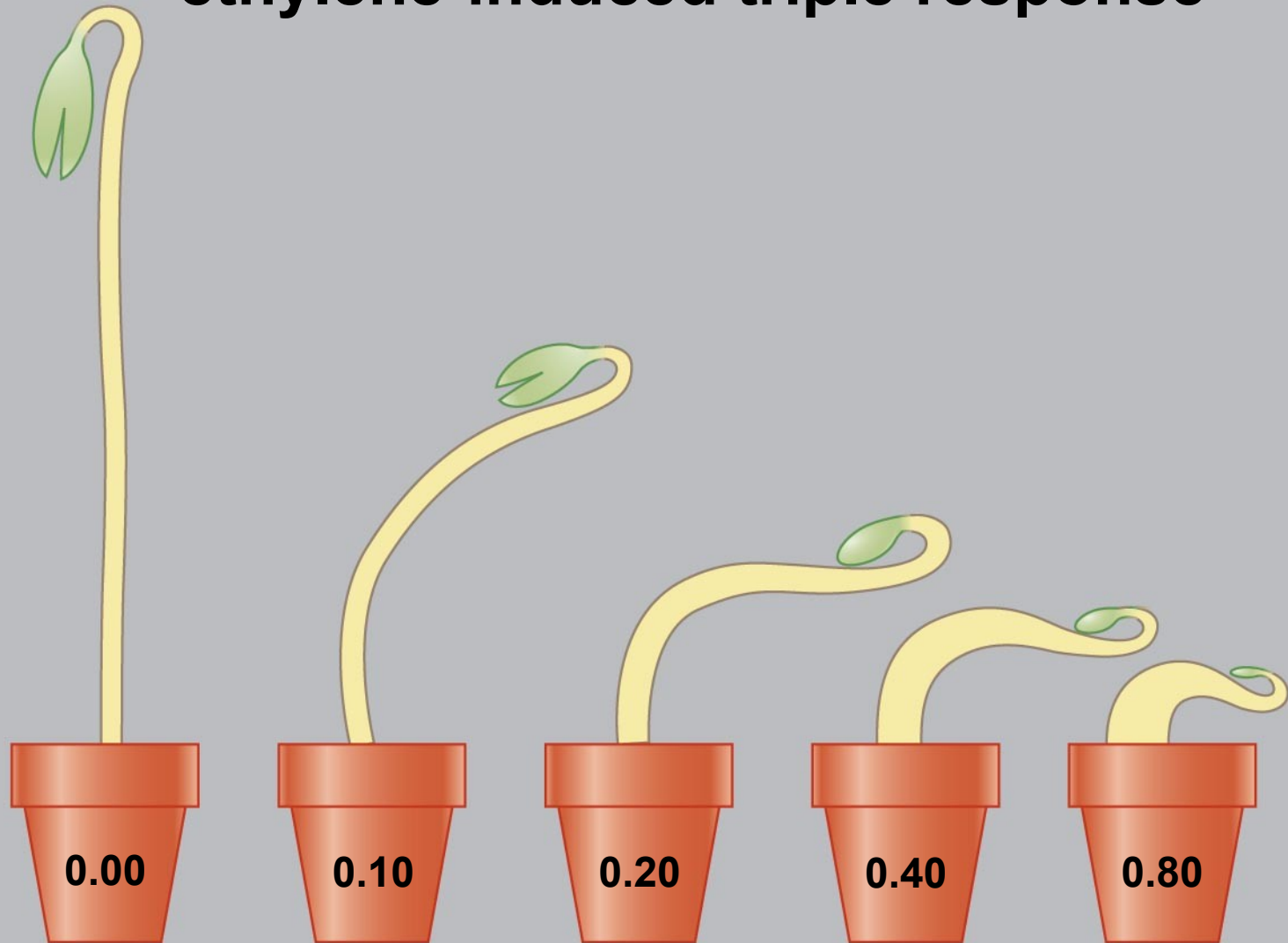
Fruit Ripening

- A burst of ethylene production in a fruit triggers the ripening process.

The Triple Response to Mechanical Stress

- Ethylene induces the **triple response**, which allows a growing shoot to avoid obstacles.
- The triple response consists of a slowing of stem elongation, a thickening of the stem, and horizontal growth.

ethylene-induced triple response



Ethylene concentration (parts per million)

Ethylene

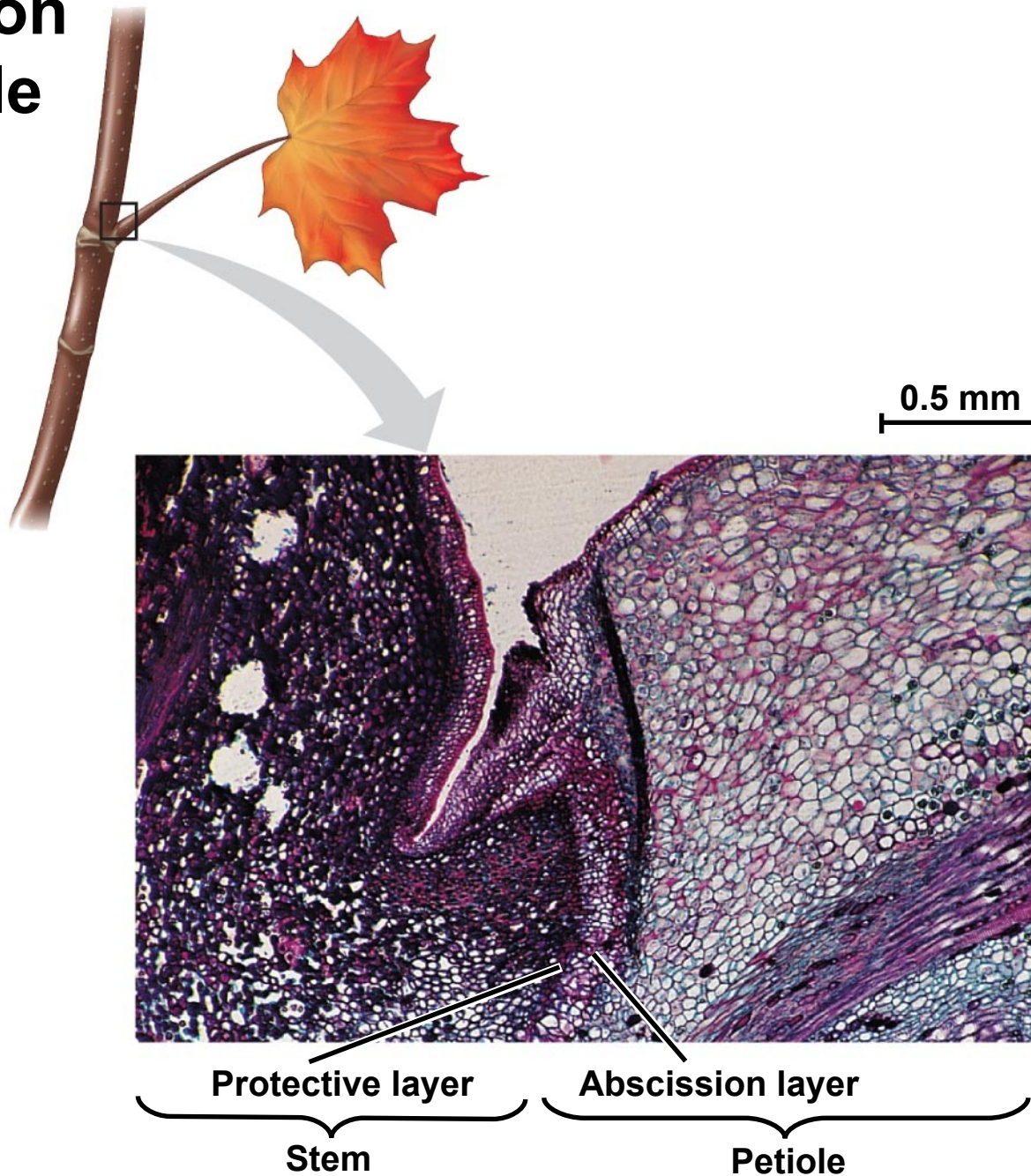
Senescence

- **Senescence** is the programmed death of plant cells or organs. A burst of ethylene is associated with **apoptosis**, the programmed destruction of cells, organs, or whole plants.

Leaf Abscission

- A change in the balance of auxin and ethylene controls leaf abscission, the process that occurs in autumn when a leaf falls.

Abscission of a maple leaf

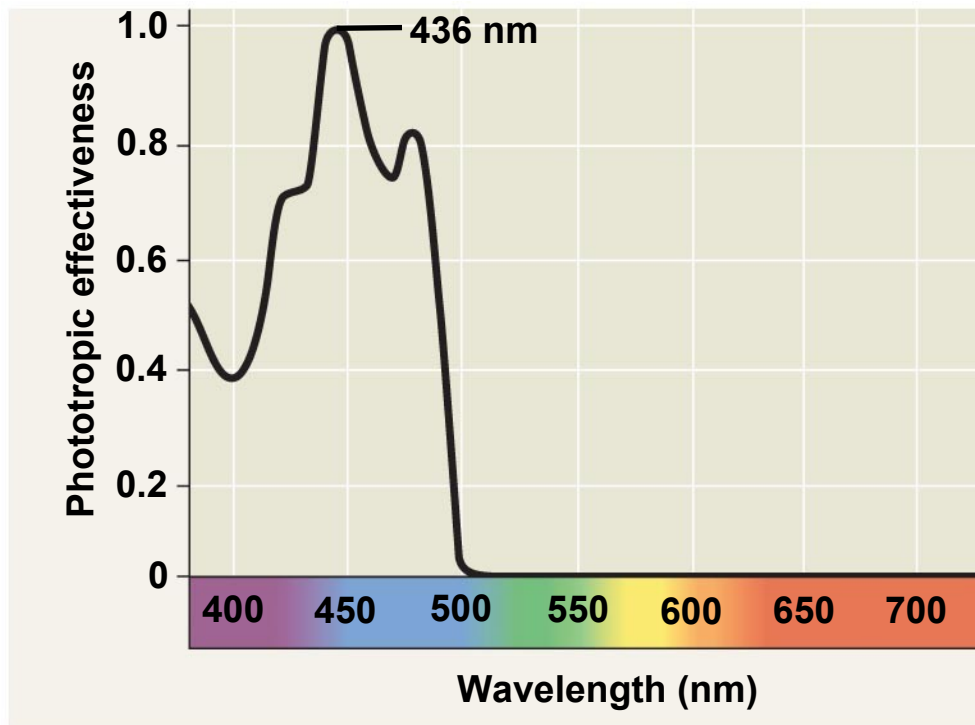


Systems Biology and Hormone Interactions

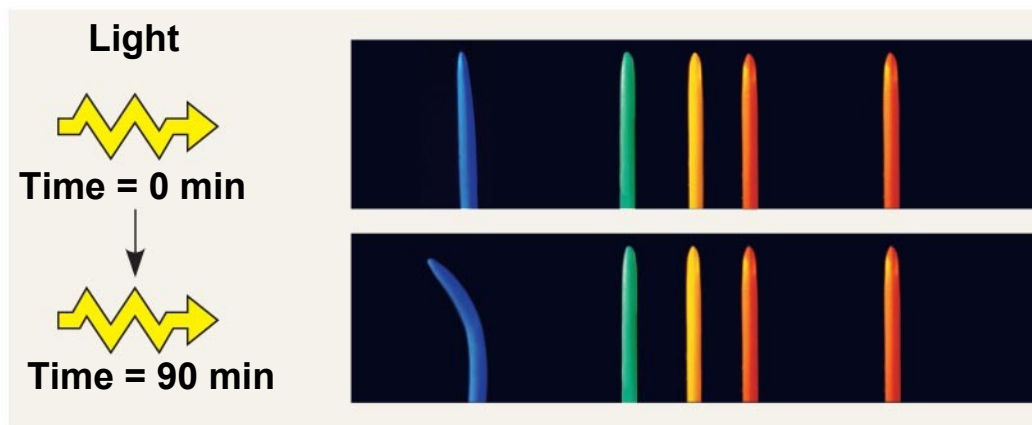
- Interactions between hormones and signal transduction pathways make it hard to predict how genetic manipulation will affect a plant.
- Systems biology seeks a comprehensive understanding that permits modeling of plant functions.

Responses to light are critical for plant success

- Light cues many key events in plant growth and development.
- Effects of light on plant morphology are called **photomorphogenesis**.
- Plants detect not only presence of light but also its direction, intensity, and wavelength (color).
- A graph called an **action spectrum** depicts relative response of a process to different wavelengths.
- Action spectra are useful in studying any process that depends on light.



(a) Action spectrum for blue-light phototropism

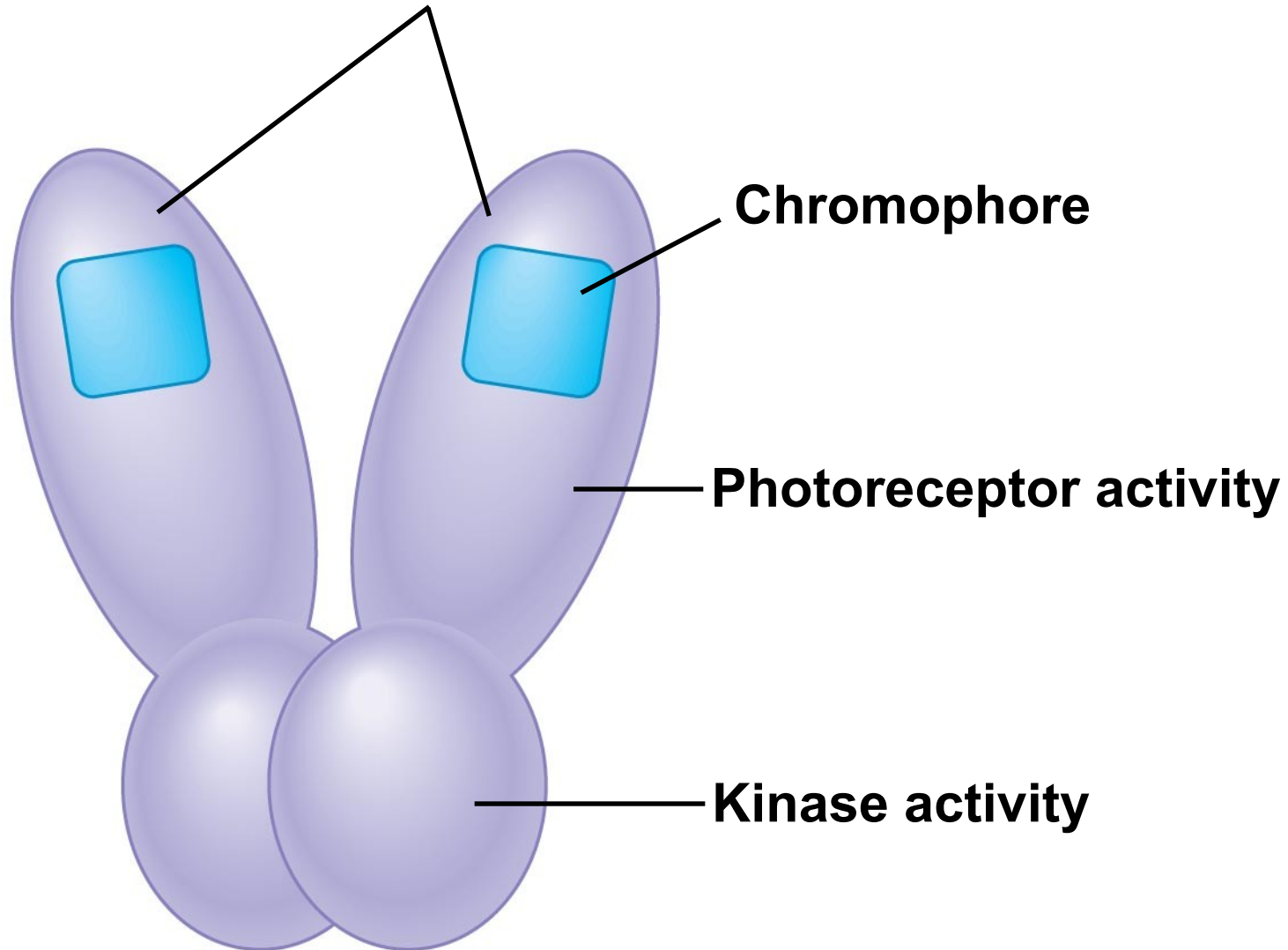


(b) Coleoptile response to light colors

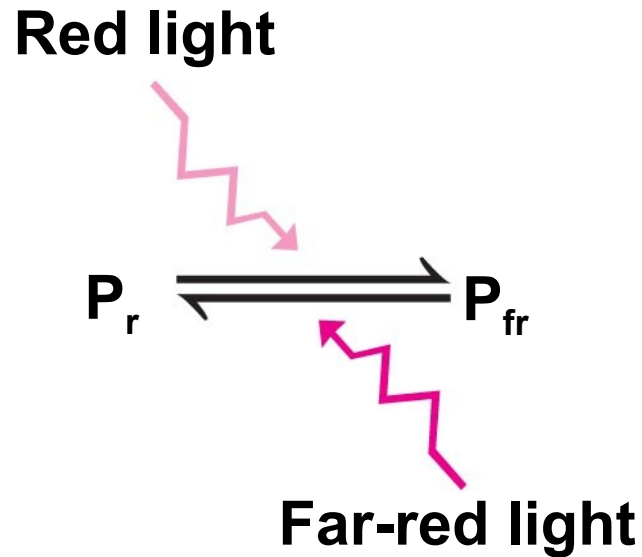
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- There are two major classes of ***light receptors***: **blue-light photoreceptors** and **phytochromes**.
 - ***Blue-light photoreceptors*** control hypocotyl elongation, stomatal opening, and phototropism.
 - ***Phytochromes*** are pigments that regulate many of a plant's responses to light throughout its life.
 - These responses include seed germination and shade avoidance.

Structure of a phytochrome

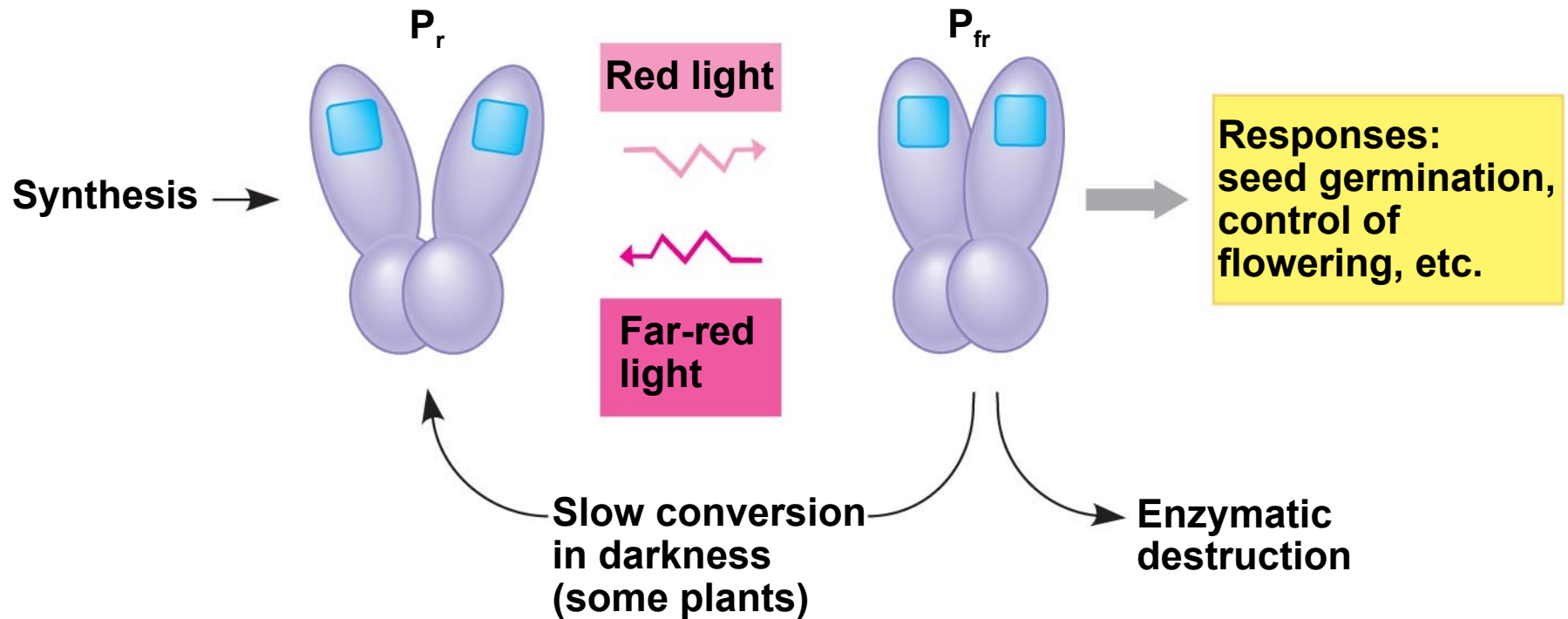
Two identical subunits



Phytochromes exist in two photoreversible states, with *conversion of P_r to P_{fr} triggering many developmental responses.*



Phytochrome: a molecular switching mechanism



Biological Clocks and Circadian Rhythms

- Many plant processes oscillate during the day.
- Many legumes lower their leaves in the evening and raise them in the morning, even when kept under constant light or dark conditions.

Sleep movements of a bean plant



Noon



Midnight

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- **Circadian rhythms** are cycles that are about 24 hours long and are governed by an internal “clock.”
 - Circadian rhythms can be entrained to exactly 24 hours by the day/night cycle.
 - The clock may depend on synthesis of a protein regulated through feedback control and may be common to all eukaryotes.
 - ***Phytochrome conversion marks sunrise and sunset, providing the biological clock with environmental cues.***

Photoperiodism and Responses to Seasons

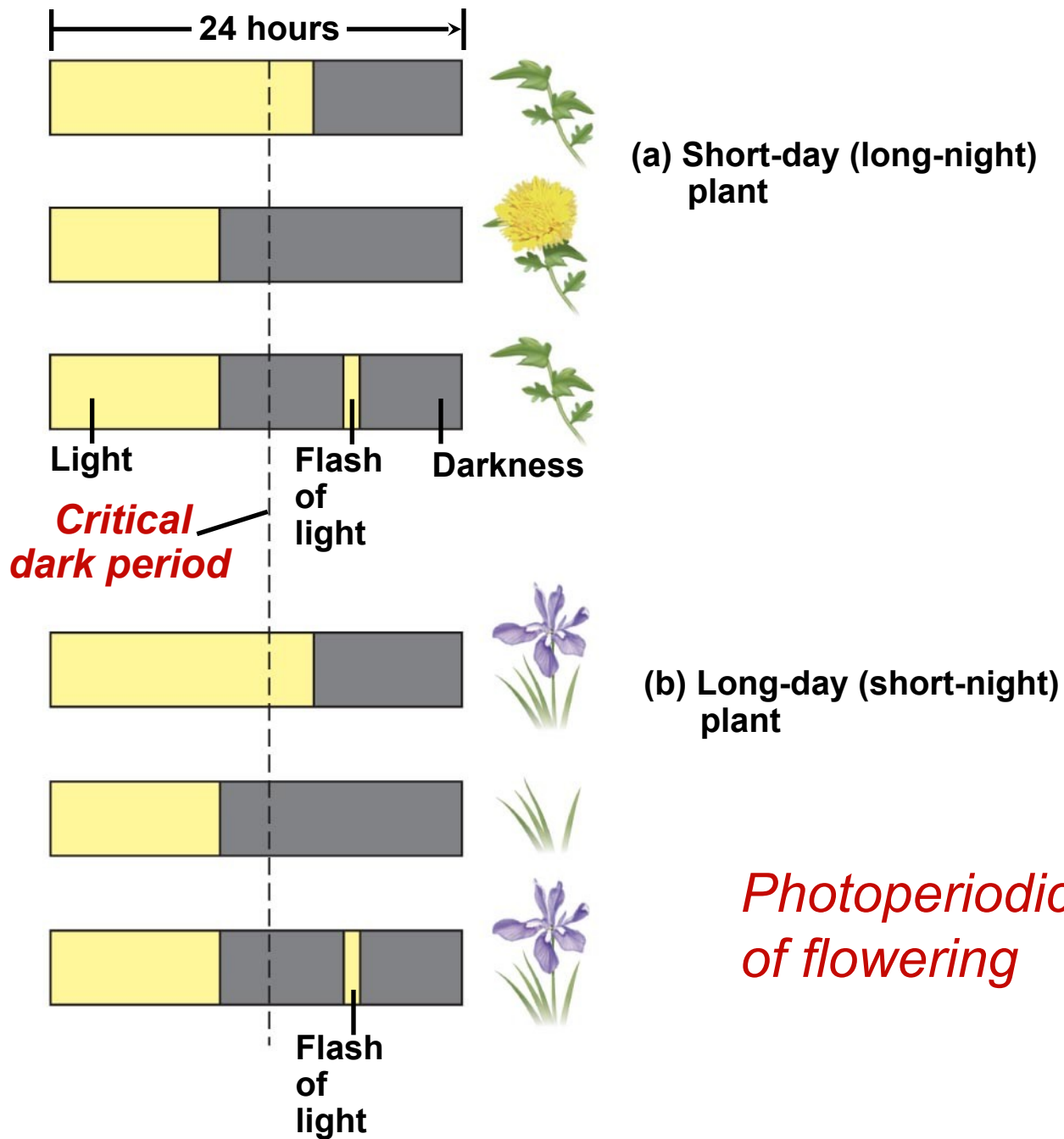
- Photoperiod, the relative lengths of night and day, is the environmental stimulus plants use most often to detect the time of year.
- **Photoperiodism** is a physiological response to photoperiod.

Photoperiodism and Control of Flowering

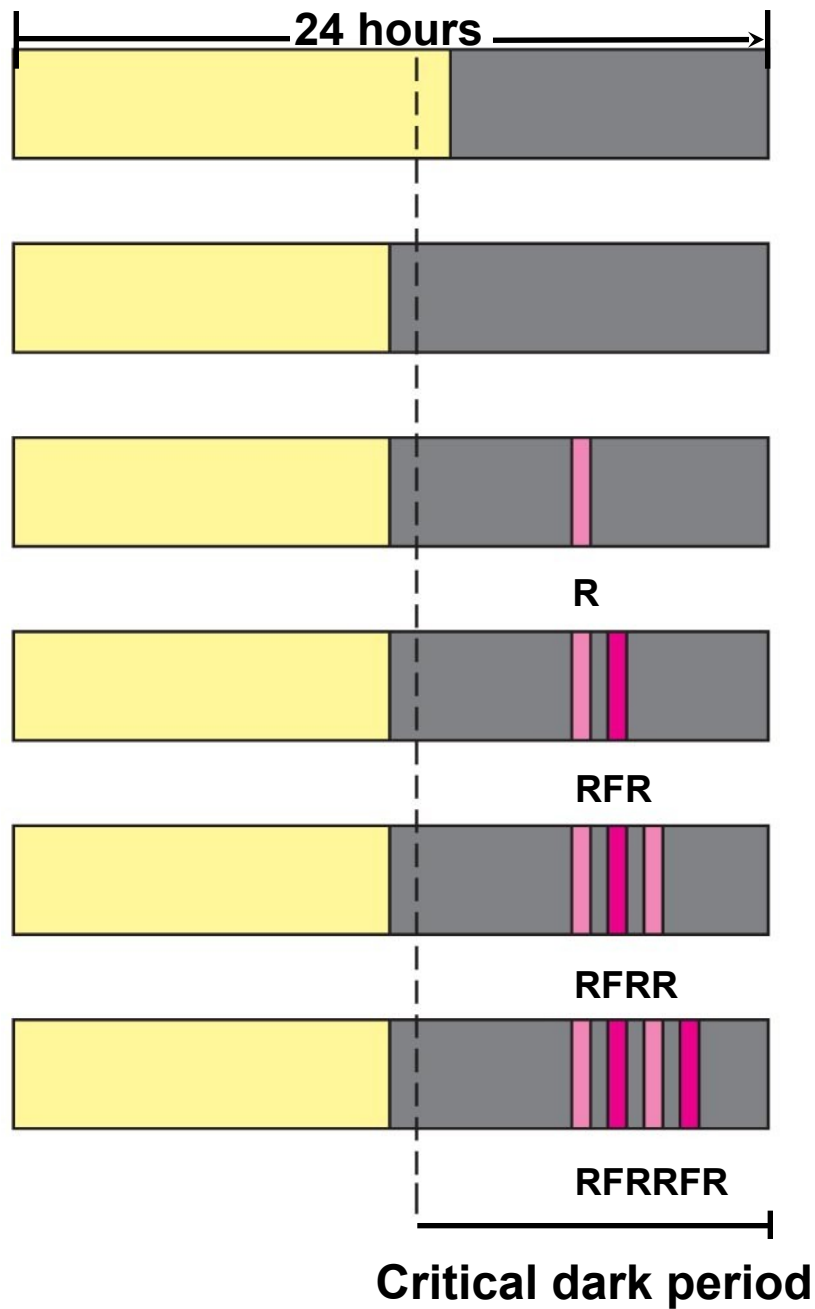
- Some processes, including flowering in many species, require a certain photoperiod.
- Plants that flower when a light period is shorter than a critical length are called **short-day plants**.
- Plants that flower when a light period is longer than a certain number of hours are called **long-day plants**.
- Flowering in **day-neutral plants** is controlled by plant maturity, not photoperiod.

Critical Night Length

- *Flowering and other responses to photoperiod are actually controlled by night length, not day length.*
- Short-day plants are governed by whether the critical night length sets a minimum number of hours of darkness.
- Long-day plants are governed by whether the critical night length sets a maximum number of hours of darkness.



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- Red light can interrupt the nighttime portion of the photoperiod.
 - Action spectra and photoreversibility experiments show that phytochrome is the pigment that receives red light.



**Short-day
(long-night)
plant**



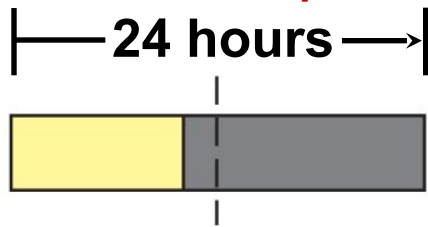
**Long-day
(short-night)
plant**

*Reversible
effects
of red
and far-red
light on
photoperiodic
response.*

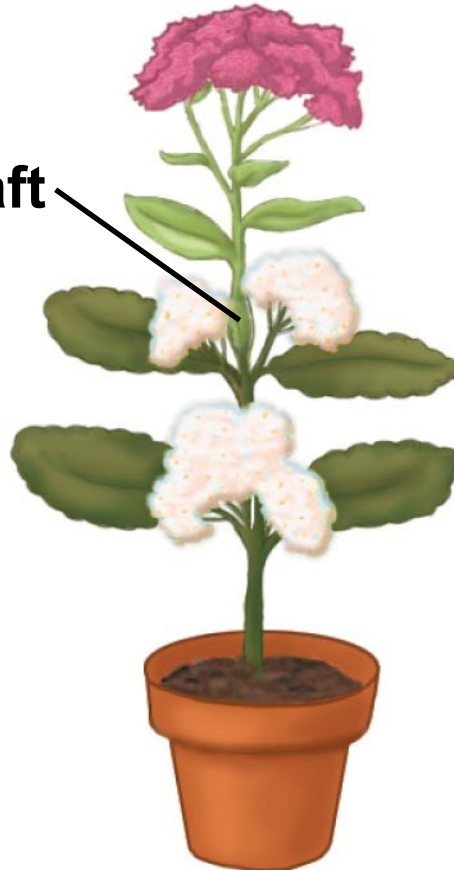
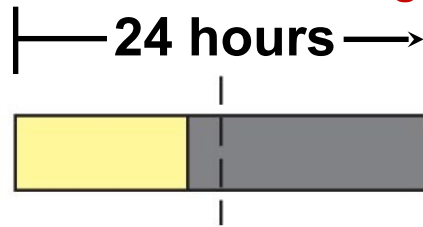
A Flowering Hormone?

- The flowering signal, not yet chemically identified, is called **florigen**.
- Florigen may be a macromolecule governed by the *CONSTANS* gene.

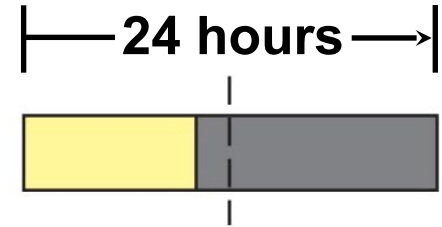
Experimental evidence for a flowering hormone



**Short-day
plant**



**Long-day plant
grafted to
short-day plant**

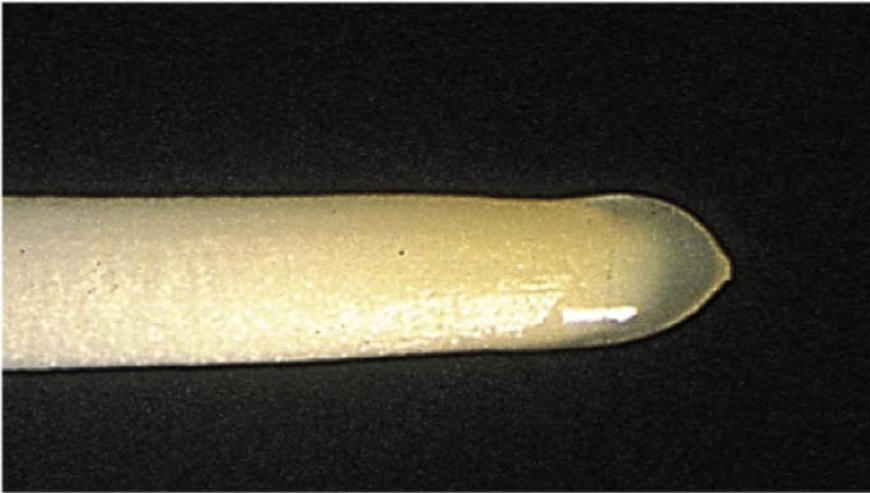


**Long-day
plant**

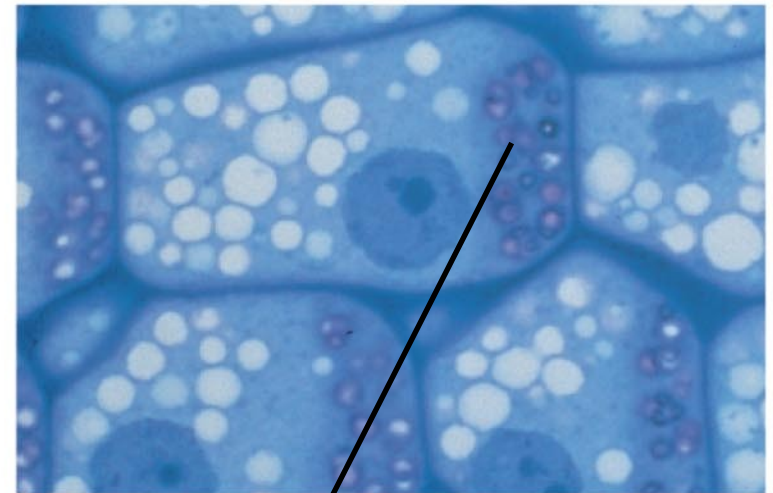
Plants respond to a wide variety of stimuli other than light

- Because of immobility, plants must adjust to a range of environmental circumstances through developmental and physiological mechanisms.
- *Response to gravity* is known as **gravitropism**.
- Roots show positive gravitropism; shoots show negative gravitropism.
- Plants may detect gravity by the settling of **statoliths**, specialized *plastids containing dense starch grains*.

Positive gravitropism in roots: the statolith hypothesis

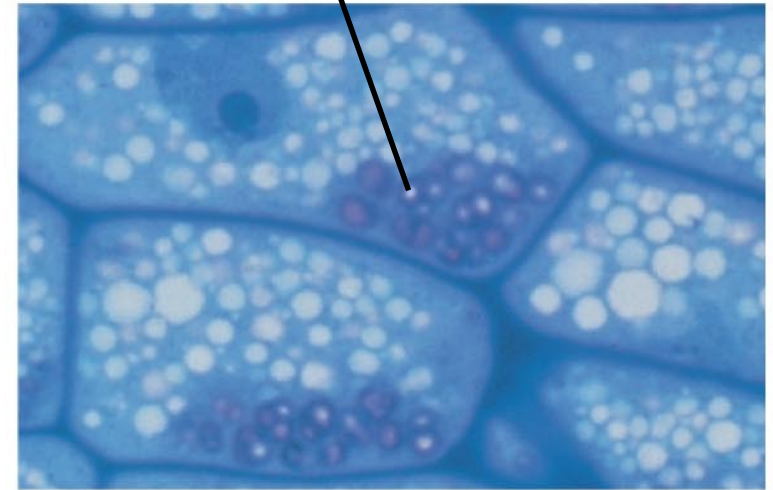


(a) Root gravitropic bending



Statoliths

20 μm



(b) Statoliths settling

Mechanical Stimuli

- The term **thigmomorphogenesis** refers to changes in form that result from mechanical disturbance. Rubbing stems of young plants a couple of times daily results in plants that are shorter than controls.
- **Thigmotropism** is growth in response to touch. It occurs in vines and other climbing plants.
- Rapid leaf movements in response to mechanical stimulation are examples of transmission of electrical impulses called **action potentials**.

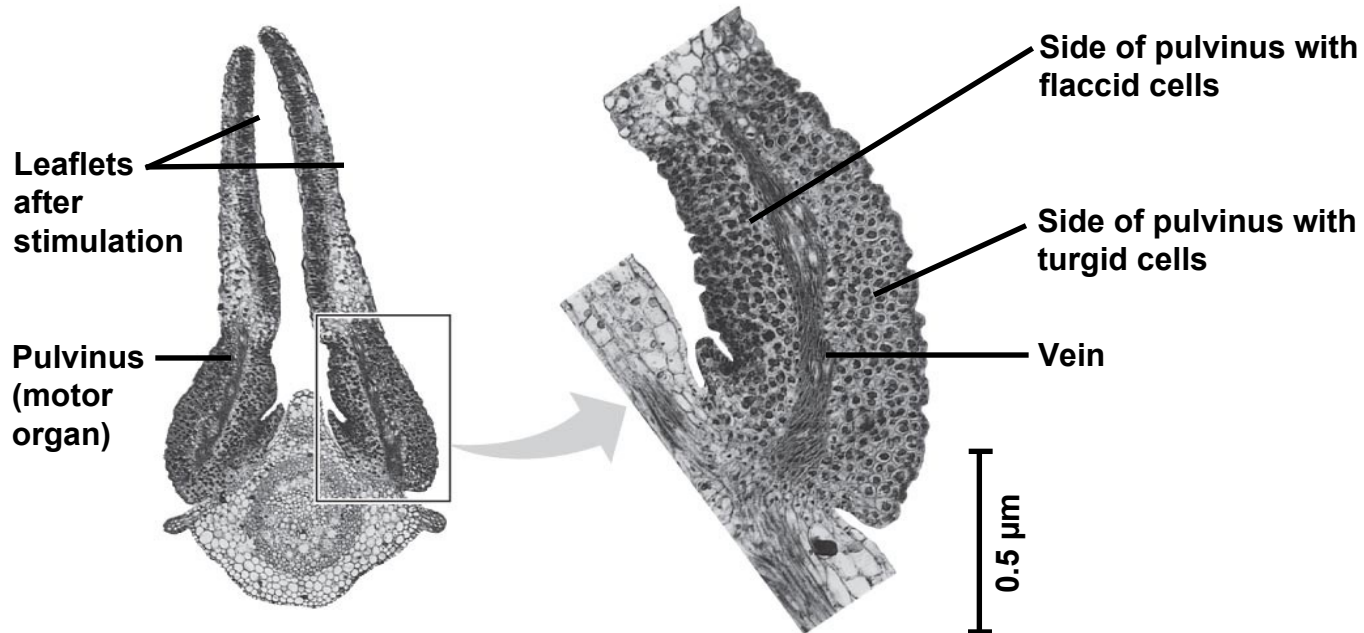
Rapid turgor movements by the sensitive plant (*Mimosa pudica*)



(a) **Unstimulated state**



(b) **Stimulated state**



(c) **Cross section of a leaflet pair in the stimulated state (LM)**

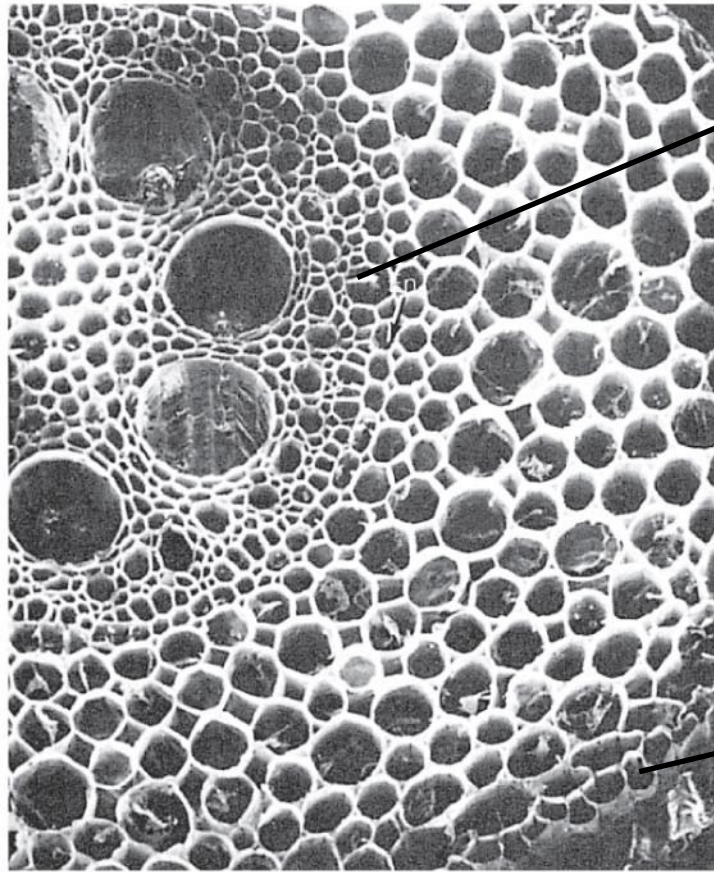
Environmental Stresses

- Environmental stresses have a potentially adverse effect on survival, growth, and reproduction.
- Stresses can be **abiotic** (nonliving) or **biotic** (living).
- Abiotic stresses include drought, flooding, salt stress, heat stress, and cold stress.

Drought and Flooding

- During drought, plants reduce transpiration by closing stomata, slowing leaf growth, and reducing exposed surface area.
- Growth of shallow roots is inhibited, while deeper roots continue to grow.
- Enzymatic destruction of root cortex cells creates air tubes that help plants survive oxygen deprivation during flooding.

A developmental response of maize roots to flooding and oxygen deprivation



100 μm

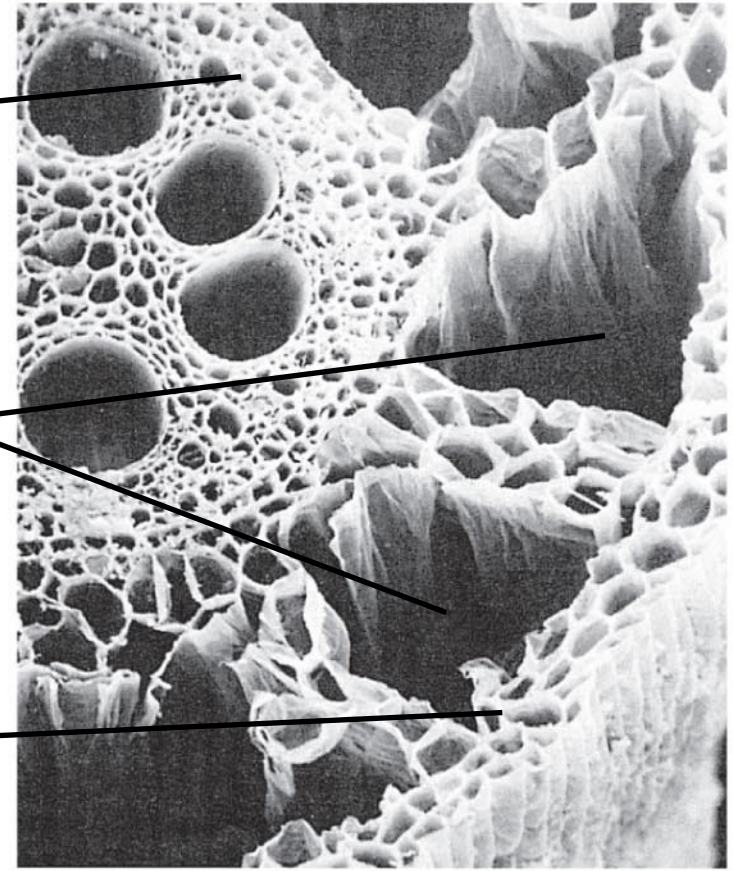
(a) Control root (aerated)

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Vascular
cylinder

Air tubes

Epidermis



100 μm

(b) Experimental root (nonaerated)

Salt Stress

- Salt can lower the water potential of the soil solution and reduce water uptake.
- Plants respond to salt stress by producing solutes tolerated at high concentrations.
- This process keeps the water potential of cells more negative than that of the soil solution.

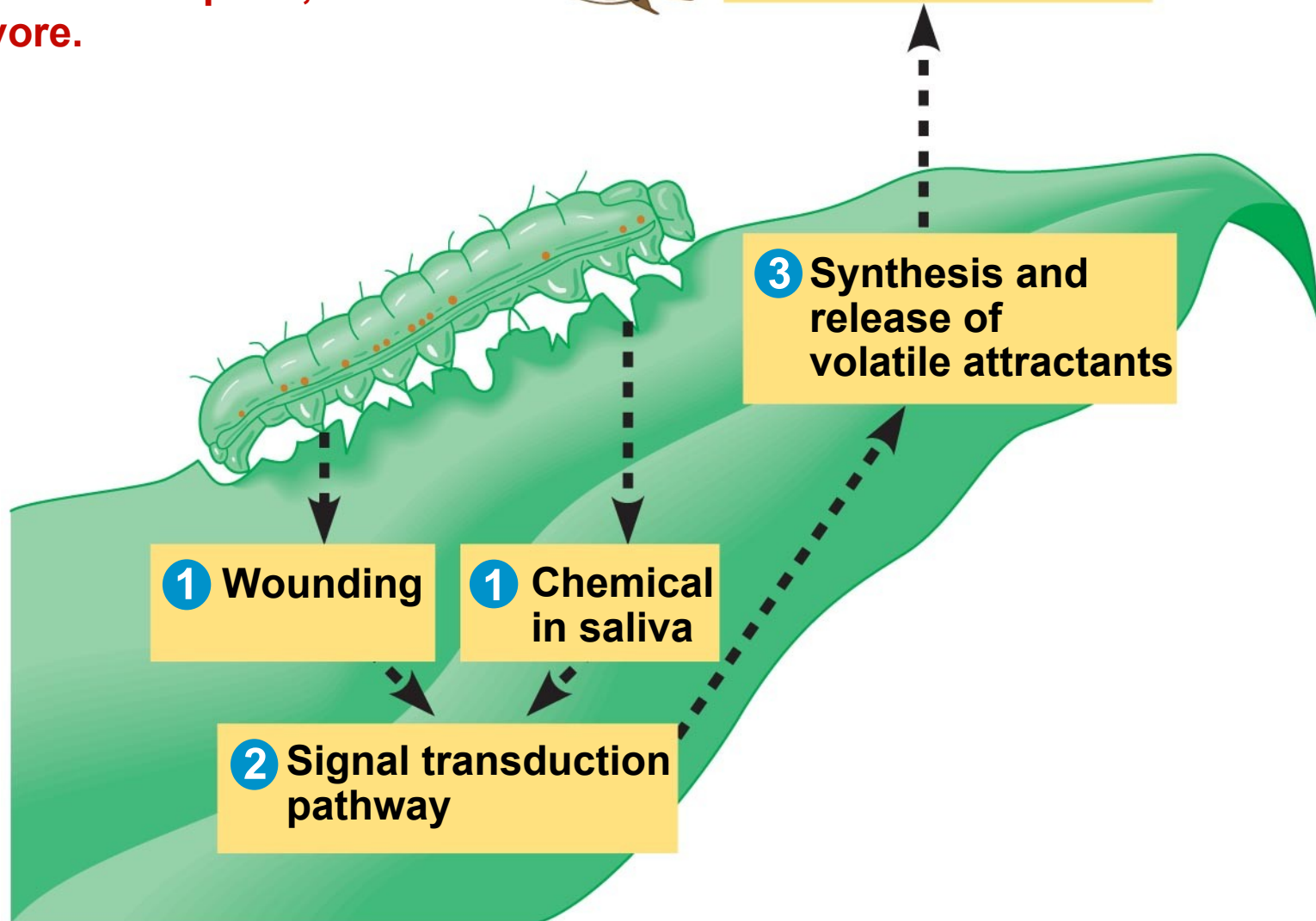
Heat Stress and Cold Stress

- Excessive heat can denature a plant's enzymes. **Heat-shock proteins** help protect other proteins from heat stress.
- Cold temperatures decrease membrane fluidity.
- Altering lipid composition of membranes is a response to cold stress.
- Freezing causes ice to form in a plant's cell walls and intercellular spaces.

Plants respond to attacks by herbivores and pathogens

- Plants use defense systems to deter herbivory, prevent infection, and combat pathogens.
- Herbivory, animals eating plants, is a stress that plants face in any ecosystem.
- Plants counter excessive herbivory with physical defenses such as thorns and chemical defenses such as distasteful or toxic compounds.
- Some plants even “recruit” predatory animals that help defend against specific herbivores.

A maize leaf “recruiting” a parasitoid wasp as a defensive response to an armyworm caterpillar, an herbivore.

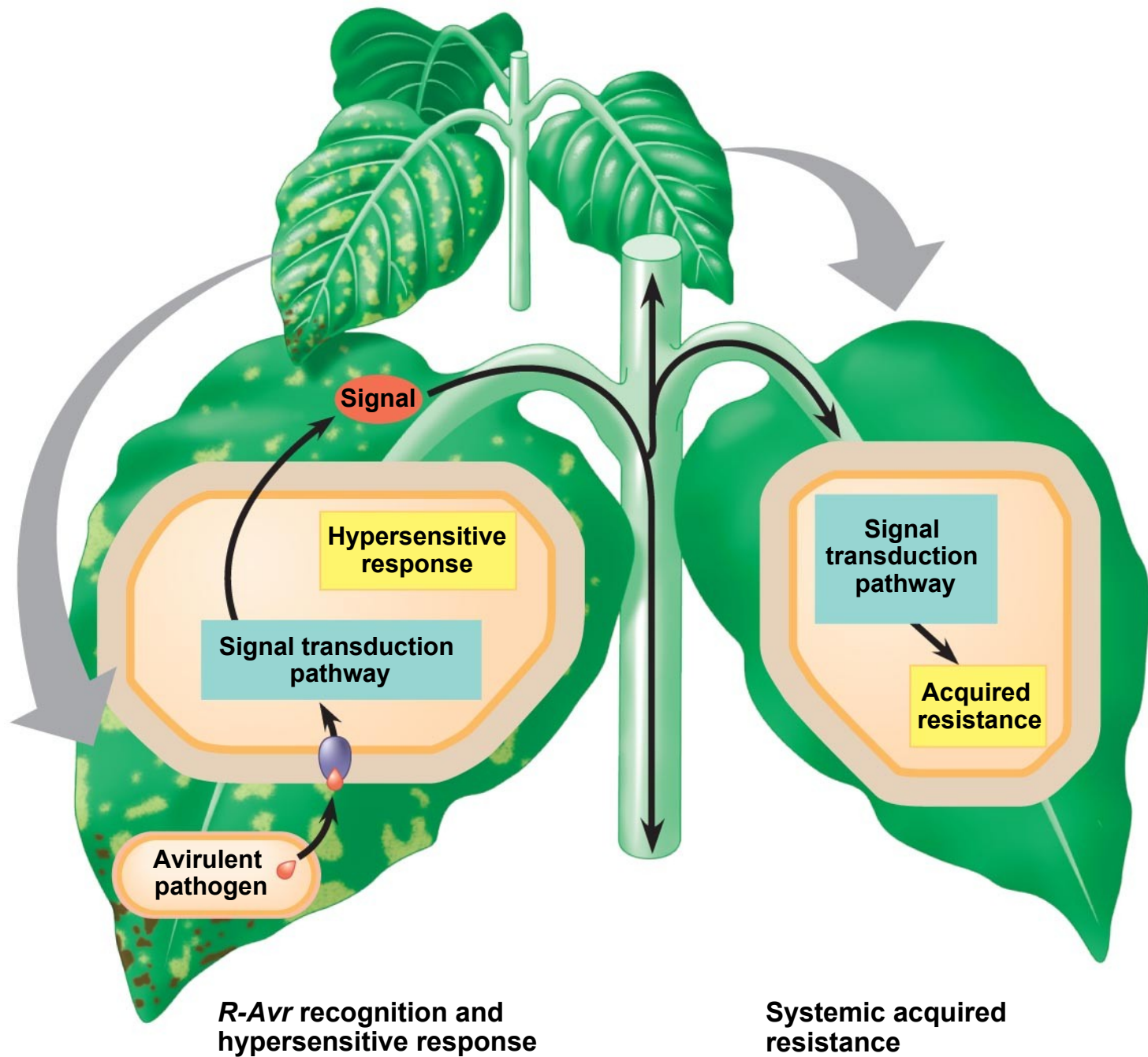


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- Plants damaged by insects can release volatile chemicals to warn other plants of the same species.
 - **Methyljasmonic acid** can activate the expression of genes involved in plant defenses.
 - A plant's first line of defense against infection is the epidermis and periderm. If a pathogen penetrates the dermal tissue, the second line of defense is a chemical attack that kills the pathogen and prevents its spread.

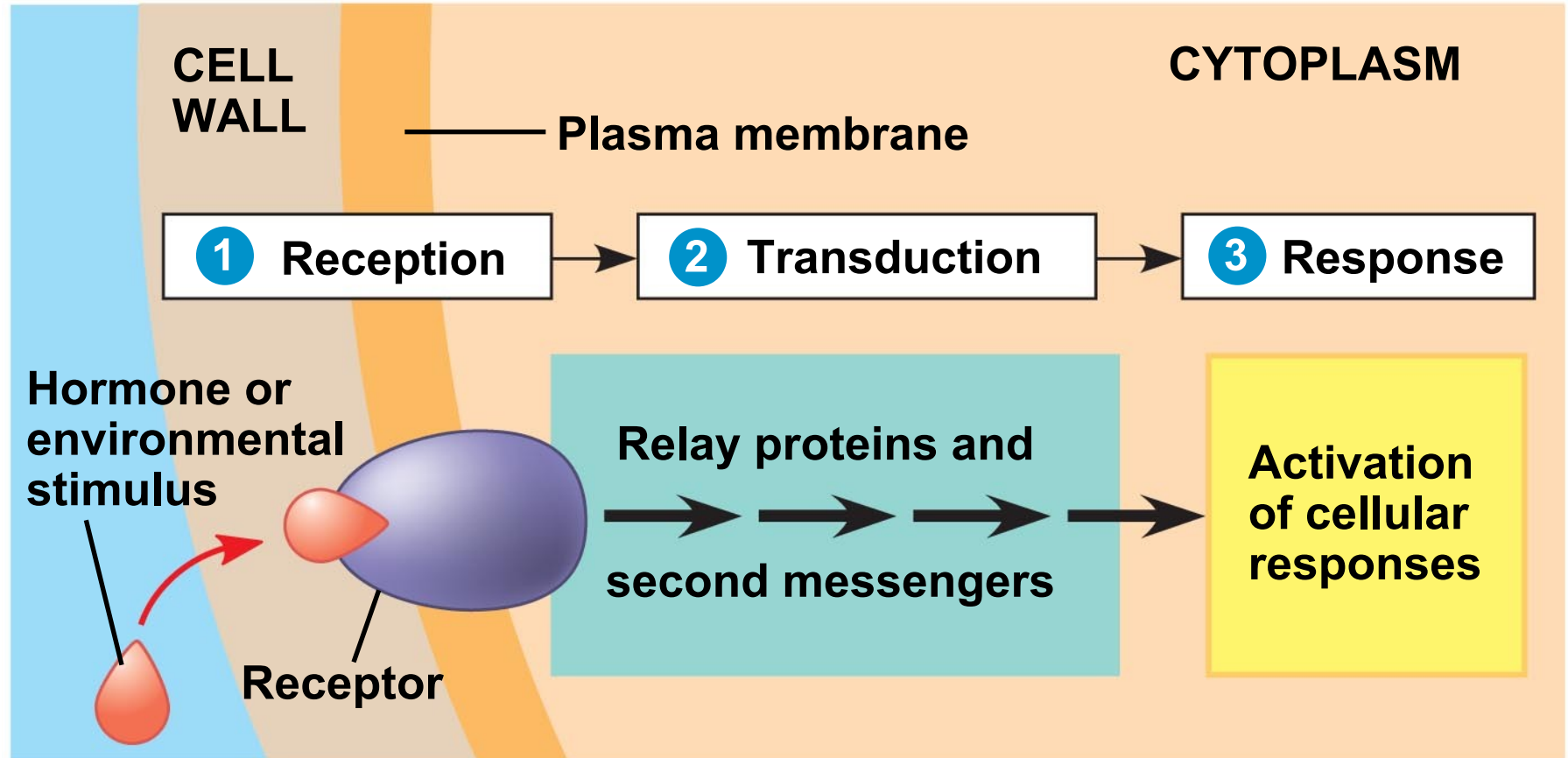
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- A **virulent** pathogen is one that a plant has little specific defense against. An **avirulent** pathogen is one that may harm but does not kill the host plant.
 - **Gene-for-gene recognition** involves recognition of pathogen-derived molecules by protein products of specific plant disease resistance (*R*) genes.
 - An *R* protein recognizes a corresponding molecule made by the pathogen's *Avr* gene.
 - *R proteins activate plant defenses by triggering signal transduction pathways.*

The Hypersensitive Response

- **The hypersensitive response**
 - Causes cell and tissue death near the infection site
 - Induces production of phytoalexins and PR proteins, which attack the pathogen
 - Stimulates changes in the cell wall that confine the pathogen.

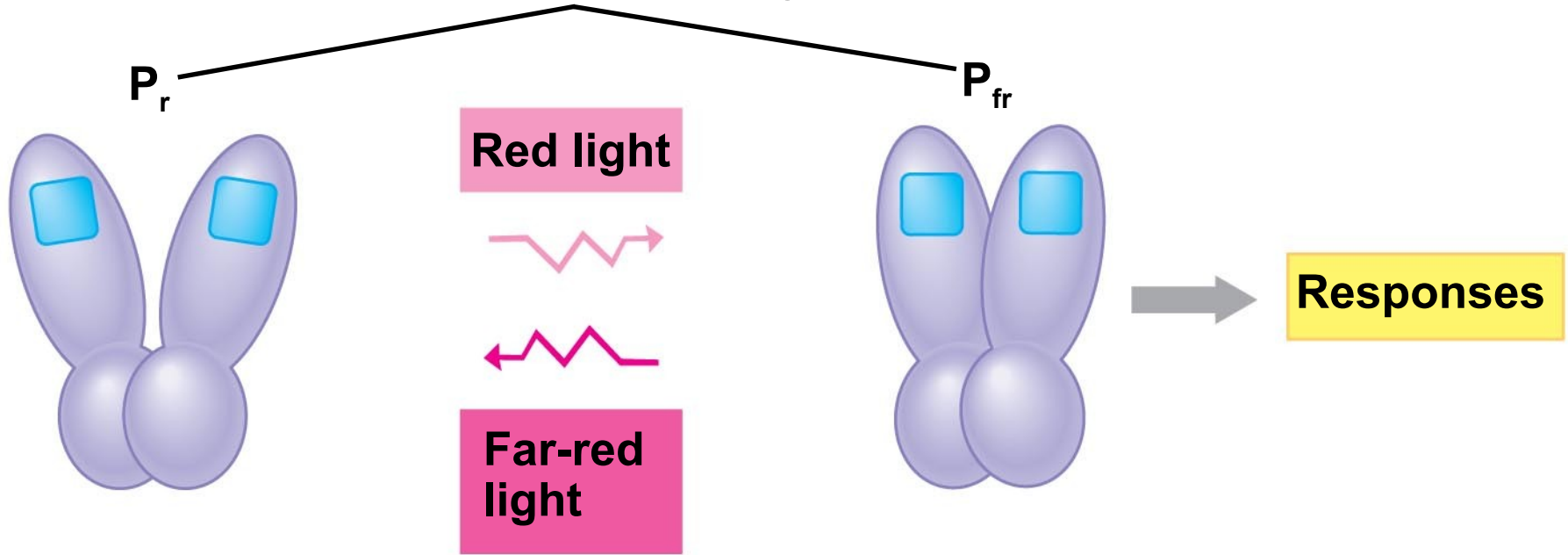


Review: Signal Transduction Pathway



Review

Photoreversible states of phytochrome



You should now be able to:

1. Compare the growth of a plant in darkness (etiolation) to the characteristics of greening (de-etiolation).
2. List six classes of plant hormones and describe their major functions.
3. Describe the phenomenon of phytochrome photoreversibility and explain its role in light-induced germination of lettuce seeds.
4. Explain how light entrains biological clocks.

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5. Distinguish between short-day, long-day, and day-neutral plants; explain why the names are misleading.
 6. Distinguish between gravitropism, thigmotropism, and thigmomorphogenesis.
 7. Describe the challenges posed by, and the responses of plants to, drought, flooding, salt stress, heat stress, and cold stress.